Learning for the Decision: Replicating and Extending End-to-End Markowitz Learning in China's A-Share Market

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Abstract

Portfolio construction usually separates return prediction from optimization: a model is fit to minimize mean squared error and its forecasts are then passed to an optimizer. Wang, Gao, Harvey, Liu and Tao (2026) argue that this separation is costly and propose end-to-end (E2E) training, in which the optimizer is a differentiable layer and the network is trained on the realized portfolio objective. We replicate their design on the same market (daily China A-shares, 2010 to 2023) using only free data, audit the paper's internal consistency, and extend it. We derive an exact, GPU-native differentiable layer for the paper's portfolio problems that replaces generic convex-optimization differentiation, fit a family of closed-form linear and dimension-reduction models (ridge, PCR, PLS, LDA) from sufficient statistics, and propose decision-aware dimension reduction. We also show that, without tradability rules the paper does not state, much of the long-only return comes from newly listed stocks locked at their upper price limit, which cannot be bought. Preliminary results show the paper's central point in our data: the models with the best statistical accuracy do not produce the best long-only portfolios.

1 Introduction

The standard "predict, then optimize" approach treats forecast errors as equally important for every stock. A long-only investor holding the top decile, however, cares about accuracy only near the top of the ranking, and an investor facing trading costs cares about signals that do not force turnover. Wang et al. (2026) embed the portfolio problem in training so that the prediction model learns what the decision needs, and report that this raises long-only returns, survives transaction costs and traces a better efficient frontier. Our contributions are:

1. A reproducible replication on free data that covers the paper's market, frequency, universe filters, VWAP execution and industry classification (Section 2).
2. An audit of the paper: conventions it does not state but its tables imply, numbers that contradict each other, and claims the tables do not support, each with a resolution (Section 5).
3. An exact differentiable portfolio layer with an $O(N)$ closed-form Jacobian (Section 3).
4. Extensions in the prediction layer: closed-form linear and dimension-reduction models, decision-aware PCA, and mixture-of-experts models (Section 4).

2 Data and replication design

We combine two free Qlib-format dumps of Tushare data. The first provides raw daily prices, VWAP, adjustment factors, market capitalization, price limits, ST intervals, Shenwan level-1 industries and 163 point-in-time financial indicators with announcement dates, for 5,490 stocks including delisted ones. The second supplies 2007 to 2009 price history and the CSI 500 index. An independent JoinQuant copy confirms the daily VWAP returns (correlation 0.9995 over 479,000 stock-days). Table 1 compares the design with the paper's.

Table 1: Replication design.

	Wang et al. (2026)	This study
Sample	A-shares 2010–2023, 5.96M stock-days	A-shares 2010–2023, 10.04M stock-days, 2,953 stocks/day
Target	VWAP(t+1) →VWAP(t+2) minus CSI 500	Same (index leg close-to-close over day t+2)
Universe	excludes ST and suspended stocks	Same, on decision day t
Tradability	not discussed	drop a stock's first 60 trading days and days locked at the upper limit on t+1
Characteristics	415 daily, 31 Shenwan industries	47 price/volume, 163 fundamental, 10 valuation, 31 Shenwan
Windows	39m train, 9m validation, 3m test, quarterly roll	Same, 40 windows (2014Q1–2023Q4), 2-day embargo

3 Methodology

Two-stage benchmark. Stock i's market-adjusted return is modeled as $r_{i,t+1} = g(\boldsymbol{x}_{i,t};\boldsymbol{\theta}) + \epsilon_{i,t+1}$, with g linear or a feed-forward network (NN1 to NN5, layers 32-16-8-4-2), fit by $\min_{\boldsymbol{\theta}} \frac{1}{NT}\sum_{i,t}(r_{i,t+1} - g(\boldsymbol{x}_{i,t};\boldsymbol{\theta}))^2$ with Adam and early stopping. Each day the forecasts are sorted into deciles; the long-only portfolio holds the top decile.

End-to-end learning. E2E solves the bilevel problem

$$\min_{\boldsymbol{\theta}} \frac{1}{T}\sum_t \Big[-\boldsymbol{w}_t^{\star\top}\boldsymbol{r}_{t+1} + \tfrac{\gamma}{2}\|\boldsymbol{w}_t^\star - \boldsymbol{w}_{t-1}^\star\|_1\Big], \quad \boldsymbol{w}_t^\star = \arg\min_{\boldsymbol{w}\in\mathcal{W}_t} -\boldsymbol{w}^\top\hat{\boldsymbol{r}}_{t+1} + \zeta\|\boldsymbol{w}\|_2^2 + \tfrac{\gamma}{2}\|\boldsymbol{w}-\boldsymbol{w}_{t-1}^\star\|_1, \tag{1}$$

with $\hat{\boldsymbol{r}}_{t+1} = g(\boldsymbol{x}_t;\boldsymbol{\theta})$ and $\mathcal{W}_t = \{\boldsymbol{w} : \boldsymbol{0} \le \boldsymbol{w} \le u_t\boldsymbol{1},\ \boldsymbol{1}^\top\boldsymbol{w} = 1\}$, where $u_t = 1/(0.1N_t)$ so that the unregularized solution is the equal-weighted top decile. The term $\zeta\|\boldsymbol{w}\|^2$ smooths the otherwise piecewise-constant solution of the linear program, and γ is the transaction cost rate. The gradient follows the chain rule $\nabla_{\boldsymbol{\theta}} = \sum_t (\partial\hat{\boldsymbol{r}}/\partial\boldsymbol{\theta})^\top(\partial\boldsymbol{w}^\star/\partial\hat{\boldsymbol{r}})^\top\partial f_u/\partial\boldsymbol{w}^\star$.

An exact differentiable portfolio layer. The paper differentiates the lower level through its KKT conditions with a generic solver. For problem (1) this is unnecessary. The objective is separable apart from the budget constraint, so with multiplier τ each weight solves a one-dimensional problem:

$$w_i^\star = \Pi_{[0,u]}\big(S_{p_i}(\hat{r}_i - \tau)\big), \qquad S_p(a) = \begin{cases} (a-c)/2\zeta & a > 2\zeta p + c \\ (a+c)/2\zeta & a < 2\zeta p - c \\ p & \text{otherwise} \end{cases}, \quad c = \gamma/2, \tag{2}$$

where p_i is yesterday's weight and τ is the unique root of $\sum_i w_i^\star(\tau) = 1$, found by bisection. With $\gamma = 0$ this is the Euclidean projection of $\hat{\boldsymbol{r}}/2\zeta$ onto the capped simplex. Let F be the set of stocks strictly inside their bounds and off the plateau at p_i. Differentiating the budget identity gives

$$\frac{\partial\boldsymbol{w}^\star}{\partial\hat{\boldsymbol{r}}} = \frac{1}{2\zeta}\Big(\boldsymbol{D}_F - \frac{\boldsymbol{1}_F\boldsymbol{1}_F^\top}{|F|}\Big), \tag{3}$$

with $\boldsymbol{D}_F$ the diagonal indicator of F. The backward pass is therefore $O(N)$: subtract the mean upstream gradient over F and scale. We verified (2) against a general solver (60 random problems, gap below 10^{-17}) and (3) against finite differences (relative error 10^{-11}).

Evaluation. We follow conventions recovered from the paper's tables (Section 5): H–L is $(H-L)/2$; turnover is $\frac{1}{2}\|\boldsymbol{w}_t - \boldsymbol{w}_{t-1}\|_1$ and costs are γ times turnover; returns are annualized with 243 trading days; tests use Newey–West errors with $\lfloor T^{1/4}\rfloor$ lags.

4 Extensions in the prediction layer

Closed-form linear family. Every linear model predicts with one vector $\boldsymbol{\beta}$, so it can be fit from a window's sufficient statistics $\{n, \sum\boldsymbol{x}, \boldsymbol{X}^\top\boldsymbol{X}, \boldsymbol{X}^\top\boldsymbol{y}, \text{class sums}\}$, accumulated on the GPU in one pass. With $\boldsymbol{C}$ and $\boldsymbol{c}$ the centered covariances: ridge $\boldsymbol{\beta} = (\boldsymbol{C} + \lambda\bar{\sigma}^2\boldsymbol{I})^{-1}\boldsymbol{c}$; principal components regression $\boldsymbol{\beta} = \boldsymbol{V}_K(\boldsymbol{V}_K^\top\boldsymbol{C}\boldsymbol{V}_K)^{-1}\boldsymbol{V}_K^\top\boldsymbol{c}$; PLS via SIMPLS on $(\boldsymbol{C}, \boldsymbol{c})$; and Linear Discriminant Analysis with classes given by each day's return deciles, solving $\boldsymbol{S}_b\boldsymbol{v} = \lambda\boldsymbol{S}_w(\alpha)\boldsymbol{v}$ with $\boldsymbol{S}_w(\alpha) = (1-\alpha)\boldsymbol{S}_w + \alpha\,\mathrm{tr}(\boldsymbol{S}_w)/p\,\boldsymbol{I}$, then regressing on the top m directions. A "top decile versus rest" Fisher discriminant targets the long-only decision directly. Hyperparameters are chosen per window on validation rank IC, and every setting tried is reported.

Decision-aware dimension reduction. For a linear model a learned linear bottleneck adds nothing, since $(\boldsymbol{W}\boldsymbol{\gamma})^\top\boldsymbol{x}$ is still one linear score. Reduction matters only as a constraint, so we propose $\min_{\boldsymbol{\theta},\boldsymbol{W}}\ \mathcal{L}_{\mathrm{dec}}(\boldsymbol{\theta},\boldsymbol{W}) + \mu\|\boldsymbol{W} - \boldsymbol{V}_K\|_F^2$, which starts from the PCA loadings and lets the portfolio objective move them; $\mu \to \infty$ gives PCA followed by E2E. For the networks the bottleneck feeds nonlinear layers and changes the function class. A persistence-aware variant chooses directions that minimize $\boldsymbol{v}^\top\boldsymbol{C}_\Delta\boldsymbol{v}/\boldsymbol{v}^\top\boldsymbol{C}\boldsymbol{v}$, with $\boldsymbol{C}_\Delta$ the covariance of daily changes, favouring slow, low-turnover signals for the cost experiment.

Mixtures and robust losses. A Gaussian mixture fit on PCA scores gives soft memberships $\pi_k(\boldsymbol{x})$ and a mixture of linear experts $g(\boldsymbol{x}) = \sum_k \pi_k(\boldsymbol{x})\boldsymbol{x}^\top\boldsymbol{\beta}_k$, mirroring the paper's own four-asset example in which large and small firms follow different pricing rules. A linear SVR (ε-insensitive loss) and a random-Fourier-feature kernel approximation provide robust and nonlinear two-stage baselines.

5 Auditing the original paper

Table 2 lists the most consequential findings; each numeric item was recomputed from the paper's tables. The paper's four-asset example (its Table 1) reproduces exactly, as do its Table 2 averages and its cost accounting.

6 Preliminary results

Table 3 reports out-of-sample results over the 40 test quarters. Figure 1 places statistical accuracy and portfolio quality side by side. Rows for the paper's six networks and their E2E versions are added as training completes.

Four findings stand out. First, statistical and decision quality disagree. Ridge on return ranks, LDA and PLS have the highest rank ICs (0.077, 0.073, 0.071), yet the linear model trained by MSE has the best long-only Sharpe

Table 2: Selected inconsistencies in Wang et al. (2026) and how we handle them.

Finding	Resolution
Captions say stocks are sorted on "one-quarter-ahead" forecasts; the text rebalances daily	Daily prediction and rebalancing
Smoothing term $\zeta\|\boldsymbol{w}\|_2$ is unsquared and ζ is never reported	Use $\zeta\|\boldsymbol{w}\|_2^2$ (strongly convex), tune ζ on validation
"Two-sided" turnover in eq. (19) vs "one-sided" in Table 3	Cost $= \gamma \cdot \frac{1}{2}\|\Delta\boldsymbol{w}\|_1$; matches Table 3 to 0.2 pp
H–L equals $(H-L)/2$ in every cell except NN3-E2E (54.9 reported, 53.35 implied)	Report $(H-L)/2$ and raw $H-L$
Text reports 5.96M stock-days; Table A.1 implies about 11.2M	Report our own counts
$t = 1.74$ and 1.83 are marked significant at 5% (true only one-sided)	Report one- and two-sided p-values
"E2E dominates across the risk spectrum"; on gross returns it does not for $\eta \geq 2.5$	Report gross and net separately at each η
MSE-Opt uses raw MSE forecasts whose scale is shrunk toward zero	Add a validation-tuned MSE-Opt benchmark
Labels at t+2 overlap the train/validation/test boundaries	Two-day embargo at every boundary
No tradability rule: in 2017Q1, untradeable new listings and limit-locked stocks supply the whole top-decile return (+0.76% a day; −0.04% without them)	Drop a stock's first 60 trading days and days locked at the upper limit on t+1

Table 3: Out-of-sample results, 2014–2023. Top decile is the equal-weighted long-only portfolio, before costs; returns are market-adjusted and annualized.

	Rank IC	Top decile		H–L	Turnover	Excl. bottom 30%	
Model	(NW t)	Avg %	SR	SR	%	Avg %	SR
Two-stage MSE, paper's architectures							
Linear (paper)	0.067 (21.2)	31.1	1.36	5.58	32	22.8	1.03
NN1	0.062 (21.6)	31.7	1.35	6.61	28	20.7	0.91
NN2	0.063 (21.9)	29.6	1.26	6.51	30	18.4	0.81
NN3	0.061 (20.6)	29.0	1.23	6.33	28	18.1	0.80
NN4	0.063 (21.5)	30.5	1.29	6.67	29	18.7	0.82
NN5	0.060 (20.3)	26.7	1.13	5.90	27	16.2	0.71
Two-stage, closed-form linear family (ours)							
OLS (closed form)	0.066 (20.5)	31.0	1.35	5.35	33	22.2	1.00
Ridge	0.069 (20.1)	27.8	1.23	4.85	31	19.5	0.89
Ridge-rank	0.077 (23.9)	26.8	1.20	5.28	39	20.1	0.93
PCR	0.065 (19.8)	24.6	1.09	4.42	33	17.4	0.79
PLS	0.071 (21.3)	28.7	1.28	4.94	28	19.8	0.92
LDA	0.073 (21.4)	27.4	1.20	4.83	34	20.5	0.92
LDA-top	0.061 (16.2)	13.0	0.65	2.95	17	8.4	0.43

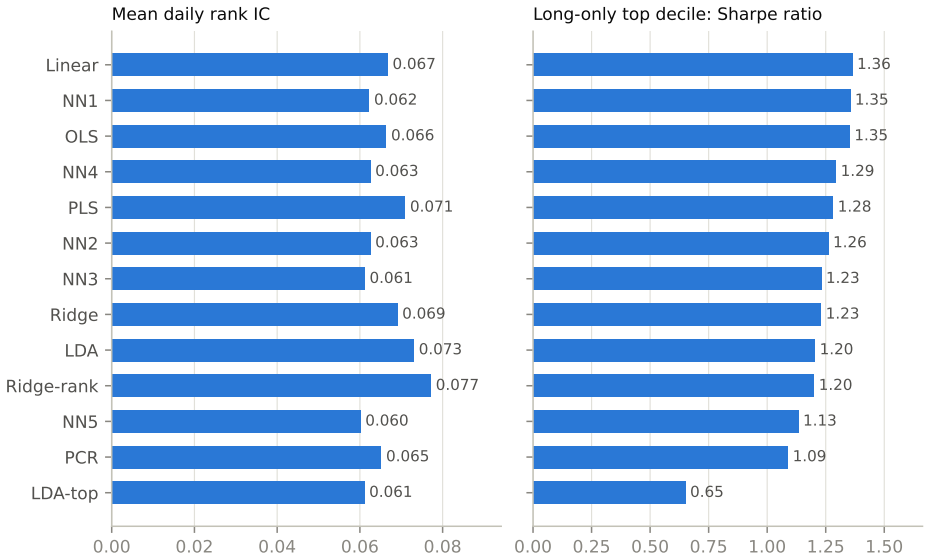
ratio (1.36). This is the paper's argument, reproduced in a different model class, and it motivates choosing the reduction by the decision rather than by variance or class separation. Second, unlike the paper, the networks do not beat the linear model in the long-only portfolio (NN3: Sharpe 1.23). Third, tradability matters: before we excluded new listings and limit-locked stocks, the same models earned over 50% a year, largely from 2015 to 2017. Excluding the bottom 30% by market capitalization, the linear model earns 22.8% (Sharpe 1.03), against the paper's 28.0% (1.44). Fourth, the CSI 500 is a mid-cap index, and the average stock in our universe beat it by 2.4% a year, so market-adjusted returns are partly a size effect.

7 Conclusion and next steps

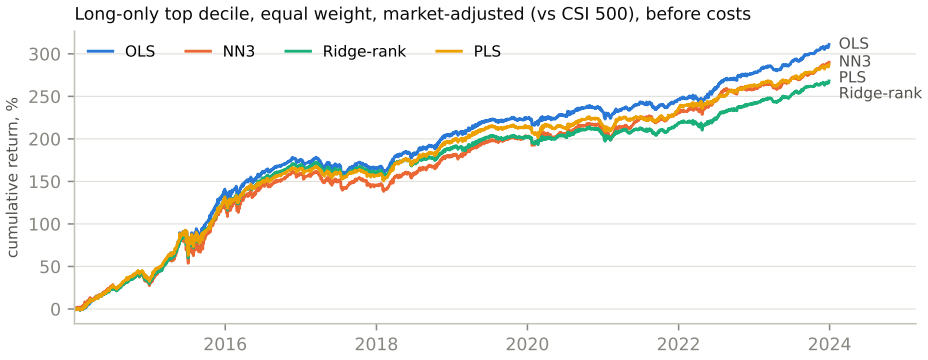
We have a validated, free-data replication of the paper's setting with explicit tradability rules, exact differentiable portfolio layers for the top-decile, cost-aware and mean–SD problems, and a closed-form linear family. All extensions are implemented and tested; the remaining step is running them at full scale: the E2E versions of the paper's networks and of our models, the transaction-cost and risk-aversion experiments, decision-aware PCA, the reduced networks and the GMM experts. Code: see the accompanying repository. Because many variants are compared, we will fix the grid in advance, report every setting and adjust for multiple testing.

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(a) Rank IC and long-only Sharpe ratio for the same models, sorted by Sharpe ratio.



(b) Cumulative market-adjusted return of the long-only top decile.

Figure 1: Statistical accuracy versus portfolio quality, 2014–2023.

64–82.

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